MS Business Intelligence

BI 5050 – Programming and Spreadsheet Modelling

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

© 2026 W. Brian Lambert, PhD, All Rights Reserved

BI 5050 – Programming and Spreadsheet Modelling
AI expectations as a student in this class

- We all need to know how to use AI to do our jobs better and faster
- In school your job is to learn – NOT to put out product
- Aggressively use AI to learn – very profitable
  - The perfect tutor, at the perfect level of detail, at the perfect time
  - Will learn things faster than humans throughout history
  - Use it to explain snippets of code to understand what the code is doing!
- <u>Don't use AI to create your models</u>
  - The course exam will be on paper with no computer

© 2026 W. Brian Lambert, PhD, All Rights Reserved

BI 5050 – Programming and Spreadsheet Modelling Overview

- The course is NOT about mastering a collection of software tools
- It's about learning how to combine programming and spreadsheet modeling to produce professional analytical products.
- The end-state goal of the course is to be able to:
  - Select the proper tools for the task at hand
  - Exchange data between tools to effectively conduct analysis
  - Produce analytical results that inform decision-making
  - Create professional, management-ready spreadsheets that facilitate the discussion of analyses and decisions

© 2026 W. Brian Lambert, PhD, All Rights Reserved

BI 5050 – Programming and Spreadsheet Modelling
This is **NOT** a software certification course

- We will use selected capabilities from each tool/library as needed to support the analytical workflow:
  - Excel → spreadsheet modeling, analysis, management-ready products
  - VBA → spreadsheet programming and automation
  - Power Query → data import, cleaning, transformation, and wrangling
  - Power Pivot → data modeling and relationships
  - Python → programming and analytical computation
  - Jupyter lab → coding IDE and documenting
  - Anaconda → manage the Python environment/packages
  - pandas, openpyxl, Seaborn → selected tools for data manipulation, interaction between python and Excel, and basic exploratory data analysis (EDA)

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Front Range Outfitters (FRO)
Decision: Retail Location Expansion

- Front Range Outfitters (FRO) is a Colorado-based outdoor recreation retailer selling hiking, camping, cycling, climbing, and seasonal outdoor products. The company currently operates several retail stores along Colorado's Front Range and has grown steadily over the past several years.

- FRO's leadership believes there is an opportunity to open one additional retail location, but the company has limited capital and wants a more rigorous analysis before committing to the expansion.

- Decision: Should FRO open a new retail location and where?

© 2026 W. Brian Lambert, PhD, All Rights Reserved

BI 5050 – Programming and Spreadsheet Modelling Topics by week

Week	Theme	Topics
1	Spreadsheet Modeling	Spreadsheet engineering and expectations; mathematical modeling, build basic Excel financial models
2	Programming	Model design, programming concepts; program design; use python and Jupyter to build repeatable processes
3	Data Wrangling and Data Modeling	Clean & tidy data; common data problems; Excel tables; Power Query fundamentals; data-wrangling workflow
4	Programming and Spreadsheet Integration	Integration of python and Excel using pandas, openpyxl and VBA
5	Exploratory Data Analysis & Uncertainty	EDA with descriptive statistics; Pivot tables; visualization; incorporating uncertainty into the analysis
6	Decision Modeling / Optimization	What-if analysis; LP model implementation with VBA automation
7	Executive Decision Meeting	Raw data → clean & tidy data → analysis → model → managerial output. Jupyter as documented analytical backend; Excel as interactive managerial frontend

© 2026 W. Brian Lambert, PhD, All Rights Reserved